MIL-C-55021A
10 December 1965
Superseding
(See Section 6)

MILITARY SPECIFICATION

CABLES; TWISTED PAIRS AND TRIPLES, INTERNAL HOOKUP, GENERAL SPECIFICATION FOR

This Specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. - This specification covers synthetic resin insulated electrical hookup wire (jacketed, unjacketed, shielded, and unshielded) in twisted pair and triple combinations both shielded and unshielded for application in internal wiring of electrical equipment at medium temperatures from -40 degrees centigrade (C) to 80 degrees C and at high temperatures from -65 degrees C to 200 degrees C.

1.2 Classification.

1.2.1 Type designation. - The type designation of cable shall be in the following form, and as specified (see 6.2):

E	C	24	P
Type of Wire (1.2.1.1)	Conductor Material (1.2.1.2)	Conductor Size (1.2.1.3)	Construction of Covering over Individual Wires (1.2.1.4)

FSC 6145

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904	903	902	SP
1st Wire (1.2.1.5)	2nd Wire (1.2.1.5)	3rd Wire (1.2.1.5)	Construction of Covering over Finished Cable (1.2.1.6)
COLOR CODE			

Type designation example: EC24P904-903-902SP

1.2.1.1 Type of wire. - The type of wire is identified by a one- or two-letter symbol as specified in table I.

Table I - Type of Wire

Type	Conductor Size	Voltage Rating
B	32 to 14	600
C	26 to 12	1000
D	24 to 1/0	3000
E	32 to 10	600
EE	32 to 8	1000

NOTE: Wire types previously identified as LW, MW, and HW are now B, C, and D respectively.

1.2.1.2 Conductor material. - The conductor material is identified by the letter "C" for copper or the letter "S" for copper clad steel.

1.2.1.3 Conductor size. - The conductor size is identified by a one or two digit number which indicates the American Wire Gage (AWG) size. (See applicable detail specification).

1.2.1.4 Construction of covering over the primary insulation of the individual wires. - The construction of the covering over the primary insulation is identified by a one or two-letter symbol which indicates the type of covering over the primary insulation.

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Symbol	Covering over insulation of wires
U	Unjacketed
B	Braided
P	Polyvinylchloride jacket
J	Polyamide jacket
T	Polytetrafluoroethylene jacket
SP	Shielded and covered with Polyvinylchloride jacket
SJ	Shielded and covered with Polyamide jacket
ST	Shielded and covered with Polytetrafluoroethylene jacket

1.2.1.5 Color code. - The color code of each individual wire is identified by a number built up from the digits designating the colors as specified in 3.3.2. The color codes for the different wires are separated by a dash. Two groups of numbers indicate a twisted pair, three groups indicate a twisted triple.

1.2.1.6 Construction of covering over the finished cable. - The covering of the finished cable is identified by a one- or two-letter symbol, following the color code, as indicated below.

Symbol	Covering over finished cable
Z	Unshielded
U	Unjacketed
P	Polyvinylchloride jacket

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Symbol(Cont'd)	Covering over finished cable(Cont'd)
J	Polyamide jacket
T	Polytetrafluoroethylene jacket
SP	Shielded and covered with Polyvinyl-chloride jacket
SJ	Shielded and covered with Polyamide jacket
ST	Shielded and covered with Polytetra-fluoroethylene jacket

2. APPLICABLE DOCUMENTS.

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

J-C-98 —— Cable and Wire, Insulated, Methods of Sampling and Testing

Military

MIL-T-5438 —— Tester; Abrasion, Electrical Cable

MIL-C-12000 —— Cable, Cord, and Wire, Electric, Packaging and Packing of

MIL-W-16878 —— Wire, Electrical, Insulated, High Temperature

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STANDARDS

Military

MIL-STD-109 ---- Quality Assurance Terms and Definitions

MIL-STD-681 ---- Identification Coding and Application of Hookup and Lead Wire

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS.

3.1 Detail requirements for individual cable types. - The individual cable requirements shall be as specified herein and in the detail specifications. In the event of any conflict between the requirements of this specification and the detail specifications, the latter shall govern.

3.1.1 Preproduction sample. - When specified (see 6.2) by the procuring activity, a preproduction sample shall be representative of the material which would normally be fabricated under normal production methods and shall be as specified herein and in the applicable detail specification.

3.2 Material. - The material shall be as specified herein. When not specifically described, the material shall be of the best quality normally used in good commercial practice. The materials used shall be virgin materials only and shall be selected, where practicable, for suitability to withstand the destructive action of salt air, moisture,

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tropical climates, and the extremes of temperature likely to be encountered.

3.2.1 Wire. - All wires used in the construction of the cable shall be of the same type (see 1.2.1.1). Individual insulated wires shall be constructed in accordance with MIL-W-16878 of the type specified in the applicable detail specification.

3.2.2 Shielding. - Shielding, when specified, shall conform to the requirements of MIL-W-16878.

3.2.3 Cable jacket. - The cable jacket shall be as specified herein or in the applicable detail specification.

3.3 Construction.

3.3.1 Covering of individual wires. - Covering of individual wires shall be as specified herein or in the applicable detail specification in accordance with MIL-W-16878.

3.3.2 Color coding. - Unless otherwise specified, (see 6.2), color coding of individual wires shall be in accordance with MIL-STD-681.

3.3.3 Wire twist. - The wires shall be grouped together and twisted so that the individual wire is not twisted and a minimum overall diameter is obtained. No fillers shall be used unless approved by the procuring activity for a particular construction.

3.3.3.1 Direction of twist. - The direction of twist may be left or right.

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3.3.3.2 Length of twist. - The length of twist shall be as specified in the applicable detail specification.

3.3.4 Covering over finished cable.

3.3.4.1 Shield. - Where construction includes a shield, it shall be applied over the wires as specified herein or in the applicable detail specification.

3.3.4.2 Jacket. - Where construction includes a jacket, it shall be applied over the wires as specified herein or in the applicable detail specification.

3.4 Electrical characteristics.

3.4.1 Dielectric strength.

3.4.1.1 Unshielded cable. - The unshielded cable shall withstand for one minute, without breakdown, the applicable voltage as specified in table II between each conductor and other conductors in sequence.

3.4.1.2 Shielded cable. - The shielded cable shall withstand for one minute, without breakdown, the applicable voltage as specified in table II between each conductor and other conductors in sequence; and between the conductors electrically connected together and the shield.

Table II - Dielectric Breakdown Voltages

Type of Wire	Conductor Size (AWG)	Volts (rms)
B	32 to 14	2,000
C	26 to 12	3,000
D	24 to 1/0	6,000
E	32 to 10	2,000
EE	32 to 8	3,000

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3.4.2 Cold bend.

3.4.2.1 Unshielded cable. - Unless otherwise specified by the procuring activity, there is no cold bend requirements for unshielded cable other than the requirements for the individual wires before twisting.

3.4.2.2 Shielded cable. - After conditioning at a temperature of -40 ± 2 degrees C, the cable shall exhibit no cracking of the insulation or other component parts when wrapped around a mandrel having a diameter 15 times the maximum diameter of the finished cable at a uniform rate, 15 revolutions per minute (rpm), for 3 complete turns. The cable shall withstand the dielectric strength requirements of 3.4.1 after bending.

3.4.3 Conductor continuity. - Each conductor shall be electrically continuous when tested in accordance with 4.5.3.8.

3.5 Environmental and service conditions. - The cable furnished under this specification shall meet the requirements of this specification when subjected to the following environmental and service conditions.

3.5.1 Spark. - All jacketed cables shall withstand, without breakdown, the spark test specified in 4.5.3.3.

3.5.2 Dielectric immersion. - The finished cable shall withstand the dielectric immersion test specified in 4.5.3.4 without insulation breakdown or damage.

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3.5.3 Heat resistance. - When the cable is subjected to the heat resistance test as specified in 4.5.3.5, shrinkage shall not be greater than 1/8 inch (3.175 millimeters). Upon completion of the heat resistance test, the cable shall meet the requirements of dielectric strength as specified in 3.4.1.

3.5.4 Abrasion resistance. - The finished cable shall withstand the abrasion resistance test as specified in 4.5.3.6 without breakdown or damage.

3.5.5 Low-temperature storage. - The cable shall be capable of operating satisfactorily after subjection to a low-temperature storage of -62 degrees C (-80 degrees F) for a period of not less than 72 hours. After exposure to this low-temperature storage test, the cable shall meet the requirements of dielectric strength as specified in 3.4.1.

3.6 Size. - The size of the cable shall conform to the requirements specified herein or as specified in the applicable detail specification.

3.7 Cable identification. - The following cable indentification shall be clearly and legibly marked at intervals of one to three feet on the outer surface of jacketed constructions:

MIL-C-55021A (and detail specification number), manufacturer's name or trade mark, type designation, and year of manufacture.

3.8 Workmanship. - Uniformity of shape and performance shall permit

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interchangeability of the cable. There shall be no kinks, splints, cuts, cracks, abrasions, or other injurious defects. Individual conductors in the cable shall be mechanically secure and electrically continuous and shall be free from any imperfections which can affect their serviceability. The cable shall be free of all foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Definitions. - The definitions of inspection terms shall be in accordance with MIL-STD-109.

4.1.2 Inspection equipment. - The Government inspector, when examining the contractor's inspection equipment, will determine that the contractor has available, and utilizes correctly, test equipment of sufficient accuracy, quality, and quantity to permit performance of the required inspections. The contractor shall have

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available a set of master gages, standards, and appropriate instruments to conduct regularly scheduled calibrations of his inspection equipment. Records of such calibrations shall be maintained and made available for Government review. The calibration of gages, standards, and instruments will be checked periodically by authorized Government personnel.

4.2 Lot size. - All cables of one type, class, size, and temper, made under the same essential conditions, produced on a substantially continuous basis, and offered for inspection at one time shall be considered a lot.

4.3 Examination. - Visual and tactile examination shall be conducted for the purpose of determining compliance with the requirements of construction, size, and workmanship.

4.4 Sample. - When a sample is required (see 6.2), it shall meet the requirements of this specification and the applicable detail specification.

4.5 Inspection.

4.5.1 Preproduction sample tests. - When required (see 6.2), the preproduction sample tests shall consist of the tests specified in table III, table IV, and the tests specified in the applicable detail specification.

Table III - Preproduction Sample Tests

Test	Requirement Paragraph	Test Paragraph
Cold bend	3.4.2	4.5.3.2

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Table III - Preproduction Sample Tests (Cont'd)

Test	Requirement Paragraph	Test Paragraph
Spark test	3.5.1	4.5.3.3
Dielectric immersion	3.5.2	4.5.3.4
Heat resistance	3.5.3	4.5.3.5
Abrasion resistance	3.5.4	4.5.3.6
Low-temperature storage	3.5.5	4.5.3.7

4.5.2 Acceptance tests. - The acceptance tests shall consist of the tests specified in table IV and as specified in the applicable detail specification. These tests shall be conducted on 100 percent of production.

Table IV - Acceptance Tests

Test	Requirement Paragraph	Test Paragraph
Dielectric strength	3.4.1	4.5.3.1
Conductor continuity	3.4.3	4.5.3.8

4.5.3 Test procedures.

4.5.3.1 Dielectric breakdown.

4.5.3.1.1 Unshielded cable. - The unshielded cable shall be tested in accordance with method 6111 of J-C-98 between each conductor and other conductors in turn at the applicable voltage specified in table II.

4.5.3.1.2 Shielded cable. - The shielded cable shall be tested in

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accordance with 4.5.3.1.1 and in addition the test voltage shall be applied between the conductors electrically connected together and the shield.

4.5.3.2 Cold bend. - Specimens of the finished cable shall have each conductor skinned, approximately 1 inch (25.4 millimeters) at each end, to the bare conductor and shall have one end attached to a mandrel and the other end weighted. After conditioning for 4 hours at a temperature of -40 degrees ±2 degrees C, and while maintained at this temperature, the cable shall be wrapped around a mandrel having a diameter 15 times the maximum diameter of the finished cable at a uniform rate of 15 revolutions-per-minute (rpm) for three complete turns. After bending and before straightening, the cable shall be subjected to the applicable dielectric breakdown test specified in 4.5.3.1.

4.5.3.3 Spark test. - The spark test shall be conducted in accordance with method 6211 of J-C-98.

4.5.3.4 Dielectric immersion. - Except for about 6 inches on each end, the cable shall be immersed in tap water containing 0.5 percent, by volume, of a suitable wetting agent. The cable shall be soaked for a minimum period of 4 hours in the water solution, which shall be maintained at 25 degrees C (77 degrees F). The wires in the cable shall then be electrically connected and subjected to a potential of 2,000 volts rms between them and the water solution and between each conductor and the shield, if applicable.

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4.5.3.5 Heat resistance. - Unless otherwise specified by the procuring activity, the specimen of finished cable shall be subjected to an air oven temperature of 80 degrees C (176 degrees F) for 96 hours. The cable specimen shall then be removed from the oven and allowed to return to room temperature. The length of exposed conductor, if any, at each end of the specimen shall be considered as shrinkage of the insulation. In no case shall the shrinkage at either end exceed the maximum value specified in 3.5.3 or in the detail specification. After 1 hour at room temperature, the cable specimen shall be wound two close turns around a mandrel having a diameter ten times the cable diameter. While in the wound position, the cable shall be subjected to the dielectric test of 4.5.3.1.

4.5.3.6 Abrasion resistance. - The abrasion resistance test shall be conducted as specified in MIL-T-5438, with the exception that 400-grit aluminum oxide shall be used in place of 4/0 garnet.

4.5.3.7 Low-temperature storage. - The cable shall be placed in a temperature chamber at a temperature of -62 degrees C (-80 degrees F) for a minimum of 72 hours. Upon completion of the 72 hour exposure, the cable shall be allowed to stabilize at normal room temperature and then shall be tested for dielectric strength as specified in 4.5.3.1.

4.5.3.8 Conductor continuity. - An ohmmeter or other suitable low-voltage device shall be used to check the continuity of each conductor.

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4.6 Preservation, packaging, packing and marking. - Preservation, packaging, packing, and marking shall be inspected to determine compliance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. - Unless otherwise specified by the procuring activity (see 6.2), the cables shall be cleaned, preserved, packaged, packed, and marked in accordance with MIL-C-12000.

6. NOTES

6.1 Intended Use. - Cables covered by this specification are used in the hookup of various electronic assemblies as specified in the detail specifications.

6.1.1 Unjacketed cable. - Unjacketed cable, polytetrafluoroethylene insulated wire, is intended for use at operating temperatures between -65 and 200 degrees C.

6.1.2 Polyvinylchloride jacketed cable. - Polyvinylchloride jacketed cable is intended for use at operating temperatures between -40 and 80 degrees C.

6.1.3 Polyamide jacketed cable. - Polyamide jacketed cable is intended for use at operating temperatures between -40 and 80 degrees C.

6.1.4 Polytetrafluoroethylene jacketed cable. - Polytetrafluoroethylene jacketed cable is intended for use at operating temperatures between -65 and 200 degrees C.

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6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Title, number and date of applicable detail specification.

c. Complete type designation. (See 1.2.1).

d. Color-coding method if other than specified (see 3.3.2).

e. Length of cable required.

f. Sample plan, when required (see 4.4).

g. Whether a preproduction sample is required. (See 3.1.1 and 4.5.1)

h. Preservation, packaging, packing, and marking method if other than specified (see 5.1).

i. Selection of applicable levels of packaging and packing. (See 5.1).

j. Levels and methods of inspection, if other than specified.

6.3 Supersession data. - This specification and the applicable detail specification supersedes the following documents:

MIL-C-55021 (Ord), dated 12 May 1958
MPD 1507, dated 11 January 1961
MPD 9171A, dated October 1961

Cross-Reference Charts

Superseded Documents	Use
MIL-C-55021 (Ord)	MIL-C-55021/1
MPD 1507	MIL-C-55021/2
MPD 9171	MIL-C-55021/3

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Superseded Wire Type	Use
LW	B
MW	C
HW	D

6.4 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MI
Navy - WP
Air Force - 11

Reviewer Activities:
Army - EL, MO, MU, MI
Air Force - 11

User Activities:
Army - WC
Air Force - 85

Preparing Activity:
Army - MI

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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